

Supplementary Material for "A multiphase equation of state of carbon addressing high pressures and temperatures", by L.X. Benedict et al. Contained herein are tables of DFT-MD and PIMC data (internal energy and pressure at different volumes/atom and temperature) for elemental carbon. Details of the calculations appear in Sections II A and II B of the main text. In the case of the PIMC results, note that statistical errors in energy and pressure values are reported as well (denoted "dE" and "dP" below).

solid DFT-MD

V (Angstroms³/atom) T (K) P (GPa) E(eV/atom)

=====			
5.39030478	2000.	40.476915	-8.51261914
5.39030478	3000.	46.749595	-8.25819642
5.39030478	4000.	53.824061	-7.98762622
5.39030478	5000.	62.498152	-7.70916727
5.39030478	6000.	68.1514	-7.4008713
5.39030478	7000.	80.271411	-7.01632481
5.11387889	2000.	71.326652	-8.44584732
5.11387889	3000.	77.902835	-8.18127692
5.11387889	4000.	85.956538	-7.91975343
5.11387889	5000.	92.622201	-7.64482137
5.11387889	6000.	100.743851	-7.33194054
5.11387889	7000.	111.059252	-6.98217392
4.86442138	2000.	105.733268	-8.33149703
4.86442138	3000.	112.544207	-8.06526169
4.86442138	4000.	118.906601	-7.80860304
4.86442138	5000.	127.231113	-7.53095907
4.86442138	6000.	135.19403	-7.2347257
4.86442138	7000.	143.734771	-6.89988759
4.86442138	8000.	156.021765	-6.46504074
4.63816923	2000.	143.345278	-8.17668507
4.63816923	3000.	150.519247	-7.91174969
4.63816923	4000.	158.794582	-7.65162315
4.63816923	5000.	166.311339	-7.37658309
4.63816923	6000.	174.428075	-7.08995641
4.63816923	7000.	183.600533	-6.76711791
4.41534817	2000.	188.869885	-7.96194005
4.41534817	3000.	196.520213	-7.70346946
4.41534817	4000.	203.093446	-7.43839908
4.41534817	5000.	212.84564	-7.16795788
4.41534817	6000.	219.916848	-6.88173518
4.41534817	7000.	228.989795	-6.56560247
4.41534817	8000.	239.55567	-6.20464621
4.24343142	2000.	231.919757	-7.74437513
4.24343142	3000.	239.034974	-7.4962372
4.24343142	4000.	247.11987	-7.23040185
4.24343142	5000.	255.612735	-6.95681674

4.24343142 6000. 262.609885 -6.67629887
4.24343142 7000. 270.935215 -6.37046381
4.07023014 2000. 282.335342 -7.48855545
4.07023014 3000. 289.369744 -7.23069884
4.07023014 4000. 297.309335 -6.9674284
4.07023014 5000. 304.017605 -6.69249134
4.07023014 6000. 313.087042 -6.41632632
4.07023014 7000. 322.162572 -6.11629308
4.07023014 8000. 328.688249 -5.78302692
4.07023014 9000. 341.072738 -5.3777581
3.91061327 2000. 335.641809 -7.19583897
3.91061327 3000. 343.189945 -6.93497945
3.91061327 4000. 350.748371 -6.67129303
3.91061327 5000. 357.141789 -6.40219278
3.91061327 6000. 368.563712 -6.12865768
3.91061327 7000. 374.470996 -5.8267245
3.76304296 2000. 392.973795 -6.87606937
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3.76304296 4000. 408.535205 -6.3492565
3.76304296 5000. 415.855986 -6.0819562
3.76304296 6000. 423.579389 -5.80606717
3.76304296 7000. 433.474706 -5.51019679
3.76304296 8000. 441.557943 -5.19128916
3.76304296 9000. 449.408939 -4.80048087
3.62620503 2000. 453.617688 -6.51477412
3.62620503 3000. 460.46887 -6.26798215
3.62620503 4000. 470.216177 -6.00024585
3.62620503 5000. 476.767756 -5.73174659
3.62620503 6000. 484.635236 -5.45934345
3.62620503 7000. 492.226751 -5.16265809
3.49896977 2000. 519.318723 -6.15806972
3.49896977 3000. 526.544353 -5.89614424
3.49896977 4000. 535.7504 -5.62540804
3.49896977 5000. 544.25838 -5.35664679
3.49896977 6000. 550.371446 -5.08237871
3.49896977 7000. 558.217641 -4.79240714
3.49896977 8000. 567.99572 -4.48088527
3.49896977 9000. 575.266072 -4.09621278
3.37692646 2000. 590.172339 -5.73222657
3.37692646 3000. 597.561705 -5.47408497
3.37692646 4000. 604.704387 -5.21646834
3.37692646 5000. 611.194345 -4.94800908
3.37692646 6000. 622.007849 -4.67275003
3.37692646 7000. 627.306291 -4.38344144
3.26952913 2000. 660.261764 -5.32488082
3.26952913 3000. 666.172392 -5.06816217
3.26952913 4000. 673.837749 -4.80966058

3.26952913	5000.	683.391296	-4.54033633
3.26952913	6000.	690.661674	-4.26641924
3.26952913	7000.	698.882894	-3.97526971
3.26952913	8000.	708.969609	-3.67082961
3.26952913	9000.	717.039641	-3.30966636
3.06832733	2000.	814.988424	-4.42384612
3.06832733	3000.	821.937311	-4.16841643
3.06832733	4000.	829.596817	-3.90008516
3.06832733	5000.	837.928593	-3.62928196
3.06832733	6000.	843.653851	-3.3605457
3.06832733	7000.	855.824425	-3.07385803
3.06832733	8000.	862.909455	-2.767147
3.06832733	9000.	871.127814	-2.40286085

liquid DFT-MD

V (Angstroms³/atom) T (K) P (GPa) E(eV/atom)

0.75	100000.	28461.766	98.401756
0.75	10000.	23571.36	70.353101
0.75	20000.	24376.7714	73.841995
0.75	30000.	24911.5044	76.73596
0.75	40000.	25416.4686	79.604654
0.75	50000.	25918.1504	82.548466
0.75	60000.	26421.0823	85.558668
0.75	70000.	26923.0649	88.639468
0.75	80000.	27431.9808	91.819332
0.75	90000.	27948.3776	95.079027
0.76	100000.	27739.9493	96.931928
0.76	10000.	22882.2085	68.793167
0.76	20000.	23706.9175	72.387772
0.76	30000.	24233.1753	75.26825
0.76	40000.	24742.4784	78.180614
0.76	50000.	25234.2161	81.108389
0.76	60000.	25724.1123	84.091876
0.76	70000.	26219.7143	87.170686
0.76	80000.	26709.7371	90.282816
0.76	90000.	27230.6451	93.609045
0.78	100000.	26397.3402	94.231856
0.78	10000.	21714.7287	66.104718
0.78	20000.	22450.6656	69.619758
0.78	30000.	22963.4338	72.500506
0.78	40000.	23458.3405	75.403401
0.78	50000.	23947.2119	78.35962
0.78	60000.	24408.7458	81.277189
0.78	70000.	24897.6922	84.384571
0.78	80000.	25378.7614	87.514942

0.78 90000. 25888.2482 90.848983
0.79 100000. 25750.9066 92.884241
0.79 10000. 21162.9904 65.031218
0.79 20000. 21862.202 68.321779
0.79 30000. 22367.416 71.188505
0.79 40000. 22851.5426 74.068107
0.79 50000. 23325.9337 76.978573
0.79 60000. 23805.3055 80.00039
0.79 70000. 24272.545 83.034978
0.79 80000. 24761.5367 86.241965
0.79 90000. 25249.5769 89.498805
0.81 100000. 24547.3894 90.378662
0.81 10000. 19974.3205 62.209501
0.81 20000. 20739.632 65.761513
0.81 30000. 21241.0733 68.654401
0.81 40000. 21705.3294 71.499651
0.81 50000. 22172.1235 74.431198
0.81 60000. 22635.117 77.429169
0.81 70000. 23102.0634 80.510853
0.81 80000. 23566.7205 83.658006
0.81 90000. 24061.4217 87.011559
0.82 100000. 23966.7869 89.134222
0.82 10000. 19634.9189 61.42816
0.82 20000. 20214.6019 64.554875
0.82 30000. 20698.6987 67.400097
0.82 40000. 21163.2538 70.27091
0.82 50000. 21624.4463 73.204438
0.82 60000. 22077.9318 76.188269
0.82 70000. 22535.9738 79.255179
0.82 80000. 23002.283 82.419381
0.82 90000. 23485.8589 85.749861
0.84 100000. 22880.7854 86.79136
0.84 10000. 18644.446 59.086075
0.84 20000. 19214.1926 62.217863
0.84 30000. 19688.4143 65.065868
0.84 40000. 20131.3503 67.875944
0.84 50000. 20578.7072 70.799589
0.84 60000. 21031.506 73.822212
0.84 70000. 21484.3774 76.913435
0.84 80000. 21938.5893 80.085465
0.84 90000. 22412.9975 83.417816
0.86 100000. 21869.2466 84.583413
0.86 10000. 17725.7824 56.833029
0.86 20000. 18287.5008 60.005477
0.86 30000. 18744.5249 62.832672
0.86 40000. 19186.272 65.684114
0.86 50000. 19620.4714 68.583409

0.86 60000. 20052.4459 71.556969
0.86 70000. 20493.576 74.6454
0.86 80000. 20946.5658 77.867595
0.86 90000. 21399.1294 81.143531
0.88 100000. 20918.6809 82.453611
0.88 10000. 16920.7791 54.83973
0.88 20000. 17415.0185 57.883956
0.88 30000. 17858.6467 60.676484
0.88 40000. 18294.2047 63.546283
0.88 50000. 18717.7299 66.440238
0.88 60000. 19145.8014 69.436256
0.88 70000. 19571.7688 72.49599
0.88 80000. 20009.8202 75.694017
0.88 90000. 20456.6908 79.001458
0.91 100000. 19612.4624 79.49584
0.91 10000. 15749.4482 51.926913
0.91 20000. 16218.1686 54.895462
0.91 30000. 16649.1402 57.69212
0.91 40000. 17061.9505 60.511854
0.91 50000. 17471.7153 63.411792
0.91 60000. 17884.0817 66.394664
0.91 70000. 18302.4407 69.491537
0.91 80000. 18727.4432 72.688676
0.91 90000. 19162.3971 76.017189
0.93 100000. 18808.3975 77.620293
0.93 10000. 15023.02 50.011223
0.93 20000. 15482.7623 53.017575
0.93 30000. 15903.148 55.803848
0.93 40000. 16306.3821 58.621372
0.93 50000. 16717.1496 61.556887
0.93 60000. 17112.5573 64.497255
0.93 70000. 17522.5684 67.603737
0.93 80000. 17940.2315 70.812354
0.93 90000. 18371.9814 74.17263
0.96 100000. 17700.3617 75.029222
0.96 10000. 14032.8591 47.305716
0.96 20000. 14471.7979 50.383687
0.96 30000. 14875.664 53.152366
0.96 40000. 15265.3351 55.954787
0.96 50000. 15657.2467 58.86159
0.96 60000. 16048.1266 61.855593
0.96 70000. 16439.6391 64.913187
0.96 80000. 16853.9658 68.189488
0.96 90000. 17266.2318 71.516199
0.99 100000. 16680.0613 72.562701
0.99 10000. 13130.2564 44.676004
0.99 20000. 13544.129 47.887032

0.99 30000. 13937.2868 50.65886
0.99 40000. 14316.4878 53.472543
0.99 50000. 14690.818 56.337563
0.99 60000. 15073.4745 59.350112
0.99 70000. 15464.1821 62.477142
0.99 80000. 15854.325 65.674829
0.99 90000. 16263.3828 69.06499
1.02 100000. 15752.5135 70.289702
1.02 10000. 12296.9371 42.50958
1.02 20000. 12705.2318 45.587687
1.02 30000. 13077.4989 48.318024
1.02 40000. 13450.6748 51.145909
1.02 50000. 13812.2175 54.009326
1.02 60000. 14185.2752 57.027064
1.02 70000. 14562.7447 60.137041
1.02 80000. 14955.7973 63.429958
1.02 90000. 15350.1902 66.798489
1.06 100000. 14632.0209 67.483693
1.06 10000. 11296.4847 39.652559
1.06 20000. 11689.7621 42.720773
1.06 30000. 12050.2719 45.457592
1.06 40000. 12405.9104 48.262942
1.06 50000. 12761.4489 51.159057
1.06 60000. 13122.1156 54.175428
1.06 70000. 13477.4636 57.254592
1.06 80000. 13853.4183 60.536211
1.06 90000. 14239.605 63.947238
1.1 100000. 13634.0575 64.933584
1.1 10000. 10411.4629 36.994206
1.1 20000. 10789.0657 40.111364
1.1 30000. 11138.1311 42.826809
1.1 40000. 11481.5351 45.638162
1.1 50000. 11819.702 48.519582
1.1 60000. 12164.7323 51.519995
1.1 70000. 12511.6648 54.626794
1.1 80000. 12876.7381 57.917214
1.1 90000. 13250.511 61.357025
1.15 100000. 12519.4052 61.988416
1.15 10000. 9429.5141 33.834863
1.15 20000. 9791.78352 37.10915
1.15 30000. 10123.5625 39.813603
1.15 40000. 10450.3874 42.610219
1.15 50000. 10780.0684 45.52322
1.15 60000. 11102.5851 48.484453
1.15 70000. 11441.1128 51.63622
1.15 80000. 11796.5173 54.967642
1.15 90000. 12150.5818 58.386097

1.21 100000. 11362.5868 58.842199
1.21 10000. 8411.75879 30.375703
1.21 20000. 8756.84984 33.865296
1.21 30000. 9074.05297 36.582027
1.21 40000. 9381.70618 39.35281
1.21 50000. 9695.65543 42.261606
1.21 60000. 10009.1062 45.271012
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1.21 80000. 10665.2897 51.752541
1.21 90000. 11009.1049 55.228448
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1.28 60000. 8916.62548 41.933052
1.28 70000. 9227.41895 45.14174
1.28 80000. 9544.65553 48.48084
1.28 90000. 9873.63566 51.98256
1.36 100000. 9108.8048 52.511296
1.36 10000. 6457.55916 24.380754
1.36 20000. 6762.16083 27.234869
1.36 30000. 7037.1947 29.891132
1.36 40000. 7312.20771 32.660399
1.36 50000. 7586.4623 35.541819
1.36 60000. 7878.67725 38.649887
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1.36 90000. 8781.7364 48.761504
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1.47 50000. 6442.78526 31.675157
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1.47 80000. 7269.77897 41.445403
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1.62 100000. 6560.51056 44.812378
1.62 10000. 4308.6428 16.377455
1.62 20000. 4550.9685 19.020409
1.62 30000. 4778.75565 21.646228
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1.62 50000. 5252.52143 27.417203
1.62 60000. 5495.84197 30.522372

1.62 70000. 5745.79943 33.79907
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1.62 90000. 6280.60675 40.974137
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1.82 10000. 3265.23993 11.971145
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1.82 80000. 4793.81203 33.127753
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2.16 100000. 3882.39386 36.22462
2.16 10000. 2146.64863 6.748585
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2.16 30000. 2487.52452 11.970261
2.16 40000. 2667.02008 14.81455
2.16 50000. 2850.73086 17.846126
2.16 60000. 3043.0043 21.11768
2.16 70000. 3243.58417 24.618852
2.16 80000. 3449.13689 28.290902
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2.88 40000. 1407.0724 8.906289
2.88 50000. 1550.23564 12.089435
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2.88 70000. 1856.51966 19.192122
2.88 80000. 2015.57678 23.059435
2.88 90000. 2180.94491 27.117642
5.39030478 100000. 877.058897 28.779962
5.39030478 10000. 146.737084 -4.58964573
5.39030478 20000. 216.413138 -1.82995048
5.39030478 30000. 287.579967 1.02008482
5.39030478 40000. 363.8458 4.20901111
5.39030478 50000. 444.23887 7.7113672
5.39030478 60000. 525.007075 11.467902
5.39030478 70000. 609.802974 15.4839644
5.39030478 80000. 696.047029 19.7168767
5.39030478 8000. 132.153596 -5.22425109
5.39030478 90000. 785.316741 24.1607722
5.39030478 9000. 140.018128 -4.88527812
4.86442138 100000. 1019.55065 28.7033835
4.86442138 10000. 207.655046 -4.24576391

4.86442138 20000. 285.71549 -1.50608302
4.86442138 30000. 366.545097 1.34185036
4.86442138 40000. 449.62666 4.46063493
4.86442138 50000. 537.560511 7.91194468
4.86442138 60000. 627.662435 11.6142983
4.86442138 70000. 723.300498 15.5934009
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4.86442138 90000. 919.091946 24.1575863
4.86442138 9000. 200.551415 -4.5691714
4.41534817 100000. 1179.13362 28.7537468
4.41534817 10000. 285.06017 -3.78934776
4.41534817 20000. 371.698217 -1.05520568
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4.41534817 40000. 552.197768 4.85664105
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4.07023014 100000. 1341.00706 28.9906181
4.07023014 10000. 370.567876 -3.26040496
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4.07023014 30000. 556.943732 2.23736623
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4.07023014 70000. 984.000317 16.1514377
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4.07023014 90000. 1217.81239 24.5144527
3.76304296 100000. 1515.1927 29.2606044
3.76304296 10000. 471.364385 -2.62508662
3.76304296 20000. 568.956792 0.0594530664
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3.76304296 40000. 780.535507 5.8788868
3.76304296 50000. 893.06541 9.19381399
3.76304296 60000. 1011.20529 12.7838692
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3.49896977 40000. 914.767012 6.51607408
3.49896977 50000. 1035.42441 9.78906763
3.49896977 60000. 1160.451 13.3261616

3.49896977 70000. 1291.71696 17.1209792
3.49896977 80000. 1426.70816 21.1253229
3.49896977 90000. 1563.29681 25.3074169
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3.26952913 10000. 712.390287 -1.13567006
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3.26952913 30000. 938.937884 4.24821284
3.26952913 40000. 1063.05166 7.24134249
3.26952913 50000. 1190.39888 10.4779532
3.26952913 60000. 1324.45228 13.9779544
3.26952913 70000. 1461.77057 17.709891
3.26952913 80000. 1603.19454 21.6426281
3.26952913 90000. 1751.67774 25.8106376
3.06832733 100000. 2112.92073 30.6911318
3.06832733 10000. 850.135054 -0.292867475
3.06832733 20000. 968.546652 2.34468074
3.06832733 30000. 1092.57304 5.05998443
3.06832733 40000. 1221.6033 8.0035577
3.06832733 50000. 1358.77171 11.2429183
3.06832733 60000. 1500.18286 14.6940231
3.06832733 70000. 1645.28744 18.3765953
3.06832733 80000. 1795.0494 22.2644429
3.06832733 90000. 1955.7643 26.4378892
2.89506861 100000. 2330.96525 31.3229703
2.89506861 10000. 999.140453 0.598527534
2.89506861 20000. 1125.70134 3.22258519
2.89506861 30000. 1252.75021 5.8944033
2.89506861 40000. 1391.26362 8.83345971
2.89506861 50000. 1534.16853 12.0106224
2.89506861 60000. 1682.97765 15.4470626
2.89506861 70000. 1837.48995 19.1141094
2.89506861 80000. 1997.65149 22.997543
2.89506861 90000. 2164.5936 27.0979357
2.74561229 100000. 2548.7559 31.9555977
2.74561229 10000. 1153.70023 1.47855891
2.74561229 20000. 1284.26908 4.09613778
2.74561229 30000. 1425.21413 6.79857189
2.74561229 40000. 1565.63738 9.68427304
2.74561229 50000. 1713.03988 12.82276
2.74561229 60000. 1872.05148 16.2616101
2.74561229 70000. 2033.83471 19.8795255
2.74561229 80000. 2202.77483 23.7488806
2.74561229 90000. 2371.4826 27.7407798
2.60095562 100000. 2798.34004 32.7362803
2.60095562 10000. 1331.22879 2.47152262
2.60095562 20000. 1471.15992 5.08844351
2.60095562 30000. 1613.8353 7.74608107

2.60095562	40000.	1765.87236	10.6609223
2.60095562	50000.	1922.40182	13.7814278
2.60095562	60000.	2084.34253	17.1392766
2.60095562	70000.	2258.64159	20.7773903
2.60095562	80000.	2430.25688	24.5407339
2.60095562	90000.	2612.74784	28.5535704
2.46223798	100000.	3079.14972	33.6192096
2.46223798	10000.	1537.31455	3.61031258
2.46223798	20000.	1685.39092	6.20868907
2.46223798	30000.	1837.59547	8.87287543
2.46223798	40000.	1995.92212	11.7484429
2.46223798	50000.	2155.97968	14.8081254
2.46223798	60000.	2330.49125	18.1834096
2.46223798	70000.	2506.94539	21.7249074
2.46223798	80000.	2693.00887	25.5054405
2.46223798	90000.	2881.70917	29.454631
2.34636796	100000.	3353.05531	34.5180164
2.34636796	10000.	1741.3688	4.68502563
2.34636796	20000.	1898.44901	7.2841801
2.34636796	30000.	2056.66468	9.93605985
2.34636796	40000.	2221.03213	12.7999997
2.34636796	50000.	2391.19211	15.8693769
2.34636796	60000.	2568.20562	19.1587369
2.34636796	70000.	2757.58032	22.722847
2.34636796	80000.	2949.56926	26.4612794
2.34636796	90000.	3145.1536	30.3659224
2.24091322	100000.	3633.60847	35.3983317
2.24091322	10000.	1955.70415	5.80295227
2.24091322	20000.	2117.73447	8.35214036
2.24091322	30000.	2286.16124	11.037927
2.24091322	40000.	2459.08184	13.8852754
2.24091322	50000.	2637.98428	16.9472388
2.24091322	60000.	2822.69945	20.230706
2.24091322	70000.	3014.95412	23.7157487
2.24091322	80000.	3215.72808	27.4309039
2.24091322	90000.	3423.86169	31.3462295
2.14452986	100000.	3931.72516	36.3842797
2.14452986	10000.	2185.28269	6.94542711
2.14452986	20000.	2355.68817	9.51987238
2.14452986	30000.	2529.38326	12.1564336
2.14452986	40000.	2710.05042	15.006118
2.14452986	50000.	2892.28114	18.0285887
2.14452986	60000.	3087.92232	21.30844
2.14452986	70000.	3285.57301	24.752899
2.14452986	80000.	3493.25397	28.4332073
2.14452986	90000.	3712.43046	32.3497259
2.05609564	100000.	4237.90311	37.3696516

2.05609564	10000.	2424.84564	8.11803398
2.05609564	20000.	2604.74619	10.6921053
2.05609564	30000.	2790.547	13.3689282
2.05609564	40000.	2975.29296	16.1684942
2.05609564	50000.	3166.80637	19.2021975
2.05609564	60000.	3367.10975	22.4425191
2.05609564	70000.	3575.95814	25.9048085
2.05609564	80000.	3785.49555	29.5038014
2.05609564	90000.	4011.30736	33.3740246

liquid PIMC

V (Angstroms^3/atom) T (K) P (GPa) dP (GPa) E(eV/atom) dE(eV/atom)

199.441277	249500.983	55.382	1.064	-664.141446	6.6112316
199.441277	499001.967	144.233	1.046	-610.816203	2.07600274
199.441277	748502.95	255.747	1.058	-203.584236	3.09075879
199.441277	1010478.98	400.007	0.938	333.536737	2.85946203
199.441277	2020957.97	941.046	0.863	1674.03414	2.03745238
199.441277	3109166.1	1474.275	0.662	2689.902	1.46941474
199.441277	5052394.91	2422.302	1.005	4475.11505	2.03971909
199.441277	10104789.8	4877.762	1.52	9069.11324	2.94903374
199.441277	14697913.7	7104.7	2.485	13224.4798	4.91732294
199.441277	49746683.1	24070.723	12.966	44900.3913	24.3677944
199.441277	161676389.	78392.187	31.899	146351.731	59.6394162
4.77787351	1010478.98	13493.472	27.992	-313.400312	0.983011247
4.77787351	2020957.97	33466.102	11.89	944.53435	0.605453294
4.77787351	5052394.91	96987.965	15.603	4078.07828	0.717700253
4.77787351	16167705.1	322840.034	209.395	14265.932	9.4355474
4.77787351	129341641.	2614001.02	1734.506	116794.76	77.6646914
4.77787351	304332393.	6163586.68	3413.763	275577.635	152.985802
4.77787351	646705556.	13082212.3	3211.441	585064.607	143.869989
4.77787351	1.03472974E+09	20931521.2	6419.486	936181.527	287.090305
2.34636796	499001.967	12953.277	58.306	-758.517329	1.25852651
2.34636796	748502.95	19791.23	80.979	-590.144624	1.70524764
2.34636796	1010478.98	27635.164	81.991	-368.324224	1.72792288
2.34636796	2020957.97	65969.748	52.453	762.291774	1.12813772
2.34636796	3109166.1	112032.554	43.41	1969.0509	1.18482992
2.34636796	5052394.91	194543.116	64.418	3921.47223	1.7165839
2.34636796	10104789.8	405562.605	162.675	8661.18851	3.8447427
2.34636796	14697913.7	594630.468	238.739	12841.0226	5.52844709
2.34636796	49746683.1	2041670.68	1145.778	44691.0678	25.0344733
2.34636796	161676389.	6653589.34	2549.2	146023.759	56.1019362
2.34636796	1.03472974E+09	42607696.9	8935.442	935806.879	197.363033
1.57846805	1010478.98	41860.555	246.693	-391.837127	3.66786871
1.57846805	2020957.97	97081.281	259.59	668.217486	3.95472023
1.57846805	3109166.1	163774.616	306.997	1840.74695	4.77673261

1.57846805	5052394.91	286609.019	265.879	3815.36371	3.93091027
1.57846805	10104789.8	601561.209	524.274	8589.4559	7.73597062
1.57846805	14697913.7	886428.024	953.794	12839.7085	14.0614827
1.57846805	49746683.1	3034642.46	3889.414	44667.0435	57.5860914
1.57846805	161676389.	9902448.07	8572.57	146192.54	126.70074
1.57846805	1.03472974E+09	63359355.	16474.359	936196.722	243.427373